INCH-POUND

MIL-E-85082/7

16 November 1992

MILITARY SPECIFICATION SHEET

ENCODER, SHAFT POSITION TO DIGITAL OPTICAL INCREMENTAL, HOLLOW SHAFT TYPE 40SWM2880R1/8A

This specification is approved for use by all Departments and Agencies of the Department of Defense. The requirements for acquiring the Encoders, Shaft Angle to Digital, described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) cited in the solicitation: MIL-E-85082.

TABLE I. Requirements.

Specifications	Value	Unit	Tolerance
Code	Sector scan mode		
Capacity	2880	trans/rev	
Resolution	0.125	degree	----
Starting torque:			
@ 25°C	12.0	oz-in	maximum
@ 54°C	12.0	oz-in	maximum
Running torque @ 25°C	11.0	oz-in	maximum
Radial play	0.0010	inches	maximum
End play	0.0010	inches	maximum
Shaft runout	0.0030	inches	maximum
Slew speed	500	rpm	maximum
Voltage	5.0	volts DC	±10%
Current	500	mA	maximum
Squarewaveform trains:			
Rise time:	< 0.5	microseconds	----
Fall time:	< 0.5	microseconds	----
Output signals:			
Logic "1"	2.4VDC @ 1.2 mA	volts	minimum
Logic "0"	0.4VDC @ 48 mA	volts	maximum
Operating speed	2.5	Hz	maximum
Operating temperature	-54° to +65°	degrees C	min.-max.
Storage temperature	-62° to +75°	degrees C	min.-max.
Life	1,000,000,000	revolutions	
Weight	4.5	lbs	maximum

AMSC - N/A

FSC 5990

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NOTES:

1. ANGLE INCREMENT SIGNAL. The angle increment signal shall go through one complete cycle in 0.125 degree ± 0.02 degree, and the symmetry (on-off ratio) shall be 50 percent ± 10 percent of the cycle width. The measurement shall be made at the 50 percent points of the output wave. The tolerance shall be non-cumulative. This signal may be continuous through the full rotation of the shaft.

2. AZIMUTH

a. Azimuth scan sector signal. The azimuth scan sector signal shall be high over an angular sector of 40 degrees. End point transitions shall not be closer than 0.02 degree to a transition of the angle increment signal. Exactly 320 transitions of the angle increment signal occur within the sector over which the azimuth scan sector signal is high.

b. Azimuth center scan pulse. The azimuth center scan pulse signal shall be 1 to 7 microseconds wide positive pulse occurring at the center of the azimuth sector scan signal. It shall occur not closer than 0.02 degree to the 160th or 161st angle increment transition which occurs when the azimuth scan sector signal is high.

c. Azimuth scan limit signal 1. This signal shall transition to the high level when the shaft is rotated clockwise 40 degrees ± 0.5 degree from the azimuth center scan position and shall remain high for all angles greater than the nominal 40 degrees out to 90 degrees.

d. Azimuth scan limit signal 2. This signal shall transition to the high level when the shaft is rotated counterclockwise 40 degrees ± 0.5 degree from the azimuth center scan position and shall remain high for all angles greater than the nominal 40 degrees out to 90 degrees.

3. ELEVATION

a. Elevation scan sector. The elevation scan sector signal shall be high over an angular sector of 10 degrees. End point transitions shall not be closer than 0.02 degree to a transition of the angle increment. Exactly 80 transitions of the angle increment signal shall occur within the sector over which the elevation scan sector signal is high. Transitions of the elevation scan sector signal shall occur when the angle increment signal is low.

b. Elevation center scan pulse. The elevation center scan pulse shall be a 1 to 7 microsecond wide positive pulse occurring at the center of the elevation scan sector signal. It shall occur not closer than 0.02 degrees to the 40th or 41st angle increment transition which occurs when the elevation scan sector signal is high.

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c. Elevation scan limit signal 1. This signal shall transition to the high level when the shaft is rotated counterclockwise 18 degrees ± 0.5 degree from the elevation center scan position, and shall remain high for all angles greater than the nominal 18 degrees out to 90 degrees.

d. Elevation scan limit signal 2. This signal shall transition to the high level when the shaft is rotated clockwise 18 degrees ± 0.5 degree from the elevation center scan position, and shall remain high for all angles greater than the nominal 18 degrees out to 90 degrees.

4. ROTATION DIRECTION SENSE SIGNAL. This signal shall be in the high state when the shaft is rotated in the clockwise direction and in the low state when rotating in the counterclockwise direction. (Direction is defined when looking at the mounting surface of the encoder.) The transitions shall occur when the direction of rotation is reversed and midway between the angle increment signal transitions. If the shaft rotation is stopped after being rotated in one direction, the signal will not change state until the shaft has been rotated in the opposite direction 0.25 degree or less.

5. Regulation and voltage shall be measured at connector.

6. Rise and fall time shall be measured with an impedance of 1 meg/47pf.

7. Use T.I. 5440 or equivalent for output signals.

8. Output high voltage with output high current flowing out of output for logic "1".

9. Output low voltage with rated fanout current flowing into output for logic "0".

10. Elevation scan sector for P/N M85082/7-02 shall be 20 degrees with exactly 160 transitions of the angle increment signal occurring within the sector over which the elevation scan sector signal is high.

11. Encoder may be mounted in any position.

12. Electromagnetic interference shall be in accordance with MIL-STD-461, Method RS03; radiated susceptibility electric field shall be 60 v/m over the frequency range 14 kHz to 10 gHz.

13. Shock shall be performed at 70 g's for 30 milliseconds.

14. Radial and end play shall be performed with a maximum load of 5 lbs.

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TABLE II. Part identifying number variant characteristics.

Part Identifying No. 1/	A ±0.030	G ±0.005	Shaft Style	Terminal End	U +0.000 -0.001	Y +0.015 -0.000	Z Maximum
M85082/7-01A	1.000	3.180	Hollow	Connector	0.875	0.690	0.562
M85082/7-02A	1.000	3.180	Hollow	Connector	0.875	0.690	0.562

1/ Part identifying number M85082/7-01A shall be the NATO standard.

1/ The suffix letter following the numerical dash number corresponds to the latest modification letter in the type designation.

TABLE III. Connector termination identification.1/

PIN NUMBER	FUNCTION
A	Angle increment
B	Scan sector
C	Center scan
D	Limit signal #1
E	Limit signal #2
F	Direction signal
G	Return
J	Return
M	+5VDC
R	Case ground

1/ Connector shall mate with MS3126E14-15S.

Custodians:
Navy - AS
Army - AR
Air Force - 85

Review activities:
DLA - ES

Preparing activity:
Navy - AS

(Project 5990-0405-7)

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